CHEAT SHEET: Topic 1 - Knowledge-Based Agent (KBA)

1. Basic Intro (KRR):

- **KRR Full Form:** Knowledge Representation and Reasoning.
- **Humans vs Machine:** Humans easily reason and interpret knowledge. Machines use KRR to store data, learn from it, and behave intelligently .
- **Reasoning:** Thinking logically to make a judgment .

2. What is KBA?

- **Definition:** Agents jo apne paas knowledge store karte hain, us par reason karte hain, update karte hain, aur fir action lete hain .

3. Architecture of KBA (The 2 Main Parts):

- **A) Knowledge Base (KB):** Agent ka "Brain Storage". Isme facts aur "sentences" store hote hain .
- **B) Inference System (IS):** Agent ka "Thinking Mechanism". Ye purane facts se *naye facts* (new sentences) derive karta hai .
    - *IS ke 2 Rules:* Forward Chaining aur Backward Chaining .
- *Working Flow:* Environment se input lo $\rightarrow$ Inference Engine check karega $\rightarrow$ KB se knowledge match karega $\rightarrow$ Action lega $\rightarrow$ KB ko update karega .

4. KBA ke 3 Main Operations:

- 🗣️ **TELL:** Agent apne KB ko batata hai ki usne environment se kya naya dekha/perceive kiya .
- ❓ **ASK:** Agent KB se poochta hai ki ab next kya action lena chahiye.
- ⚙️ **Perform:** Agent finally woh selected action perform karta hai .

5. Levels of KBA (3 Levels):

- 🧠 **Knowledge Level:** Agent ko kya pata hai aur uska goal kya hai .
- 💻 **Logical Level:** Sentences ko logic mein encode karna .
- 🏗️ **Implementation Level:** Physical actions perform karna .

6. Designing Approaches (2 Types):

- **Declarative Approach**
- **Procedural Approach**

COMPREHENSIVE CHEAT SHEET: Topic 2 - Introduction to Logic

1. The Core of Logic

In Artificial Intelligence, we use "Logic" to represent the knowledge of an agent's world and its current situation.

For any system to be officially called a "Logic", it MUST have these three well-defined parts:

- ✍️ **Syntax:** The rules for *how to write* or structure the logical statements (Formulae).
- 🧠 **Semantics:** The *meaning* and interpretation behind those written statements.
- ⚙️ **Proof Theory:** The rules used for *manipulating* or deriving new formulae.
- **Memorize this equation:** Logic = Syntax + Semantics + Proof Theory

2. What is Propositional Logic (PL)?

Propositional Logic is the simplest and most abstract form of logic, also known as **Boolean Logic**.

- **The Golden Rule of PL:** A proposition is a statement that can **ONLY be EITHER True OR False**. It can *never* be both at the same time.
- **How we represent it:** We use simple symbolic variables like A, B, P, Q to represent full sentences.
- **Examples of Propositions:** * "5 is a prime number" (True proposition).
    - "The Sun rises from the West" (False proposition).
- **What is NOT a Proposition?**
    - Questions (e.g., "How are you?"), commands, or personal opinions are *never* propositions because they don't have a strict True/False answer.

3. Three Important Vocabulary Terms

Whenever you evaluate a logical formula, it will fall into one of these three categories:

- ✅ **Tautology:** A statement that is **always TRUE**, no matter what. (Valid sentence).
- ❌ **Contradiction:** A statement that is **always FALSE**, no matter what.
- 🔄 **Contingency:** A statement that has a mix of **both True and False** possibilities depending on the situation.

4. The Syntax of Propositional Logic (2 Types)

Propositions are divided into two structural types:

- 🧱 **Atomic Propositions:** These are simple, independent sentences consisting of a single symbol. (Example: "2+2 is 4").
- 🔗 **Compound Propositions:** These are made by combining two or more Atomic propositions using parentheses () and "Logical Connectives". (Example: "It is raining today, AND the street is wet").

5. Logical Connectives (The 5 Operators)

These are the symbols used to connect simple sentences. Memorize their symbols and their strict rules:

1. 🚫 **Negation (NOT) | Symbol: $\neg$ or ~**
    - *Rule:* It reverses the truth. If P is True, $\neg P$ is False.
2. 🤝 **Conjunction (AND) | Symbol: $\land$**
    - *Example:* $P \land Q$ (Rohan is intelligent AND hardworking).
    - *Rule:* It is True **ONLY if BOTH** P and Q are True. If even one is False, the whole statement is False.
3. 🔀 **Disjunction (OR) | Symbol: $\lor$**
    - *Example:* $P \lor Q$ (Ritika is a doctor OR an engineer).

    - *Rule:* It is False **ONLY if BOTH** P and Q are False. If even one is True, the whole statement is True.
4. ➡️ **Implication (If-Then) | Symbol: $\rightarrow$**
    - *Example:* $P \rightarrow Q$ (If it is raining, then the street is wet).
    - *Rule:* It is False **ONLY in one condition:** when the first part (P) is True, but the second part (Q) is False. Otherwise, it is True.
5. ↔️ **Biconditional (If and only if) | Symbol: $\leftrightarrow$**
    - *Example:* $P \leftrightarrow Q$ (I am breathing if and only if I am alive).
    - *Rule:* It is True **ONLY if BOTH** P and Q have the exact same value (Both True OR Both False).

6. Order of Precedence (Priority)

If a formula has many connectives without brackets, solve them in this exact order (Highest to Lowest priority):

1. **Negation** ($\neg$) - *Highest Priority*
2. **Conjunction** ($\land$)
3. **Disjunction** ($\lor$)
4. **Implication** ($\rightarrow$)
5. **Biconditional** ($\leftrightarrow$) - *Lowest Priority*